

Problem

Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that: $Ax + By = D$ where D is the Greatest Common Divisor (GCD) of A and B. Given two positive integers A and B, find: $x \ y \ D = \text{gcd}(A, B)$ such that:

$$Ax + By = D$$

If multiple pairs (x, y) satisfy the equation, choose the pair with the minimum value of: $|x| + |y|$

If there is still more than one pair, choose the pair for which:

$$x \leq y$$

Input Format

A B

Constraints

$$1 \leq A, B \leq 10^9$$

Output Format

$x \ y \ D$ D is $\text{gcd}(A, B)$ $Ax + By = D$ (x, y) satisfies the minimum-selection rules.

Sample Input 0

```

20         t1=t2;
21         t2=t;
22         s1=s2;
23         s2=s;
24         findxy(b,a%b);
25     }
26     public static void main(String[] args) thro
27         BufferedReader br = new BufferedReader(
28             StringTokenizer st = new StringTokene
29             int a = Integer.parseInt(st.nextToken()
30             int b = Integer.parseInt(st.nextToken()
31             findxy(a,b);
32             System.out.println(s1 + " " + t1 + " "
33         }
34     }
```

Line: 34 Col: 2

 Upload Code as File

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☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Submissions

Leaderboard

Discussions